Barath R

Electronics & Communication Engineering Student | Embedded Systems & IoT

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Professional Summary

Third-year Electronics and Communication Engineering student specializing in embedded firmware engineering, bare-metal systems programming, and IoT hardware design. Experienced in developing close to the silicon – from register-level RISC-V trap handlers to production-grade sensor interfaces, RTOS (FreeRTOS & Zephyr) task synchronization, and cloud telemetry pipelines. Seeking an Embedded Systems Engineer internship/entry-level role to build high-reliability, real-time connected hardware products.

Education

B.E. Electronics and Communication Engineering 2024 – 2028 (Expected)
Rajalakshmi Institute of Technology, Chennai | *Currently in 3rd year* CGPA: 8.65/10

Higher Secondary Certificate (HSC) – Class XII 2023 – 2024
Montfort Matriculation Higher Secondary School, Chennai | *Tamil Nadu State Board* Percentage: 95%

Secondary School Leaving Certificate (SSLC) – Class X 2021 – 2022
Montfort Matriculation Higher Secondary School, Chennai | *Tamil Nadu State Board* Percentage: 94%

Technical Skills

- **Embedded & Firmware:** Bare-Metal MMIO, FreeRTOS, Zephyr RTOS, ESP-IDF (v6), RISC-V Assembly, Device Drivers, ISR, Timers, PWM, ADC, Low-Power Optimization
- **Hardware & Microcontrollers:** ESP32, ESP32-C6, ESP32-CAM, Raspberry Pi 5, Arduino UNO R3, Altera DE2 FPGA, Sensor Fusion (IMU, LDR, Temp)
- **PCB Design & EDA Tools:** PCB Design, KiCad, Proteus, Schematic Capture, 2/4-Layer Layout, Routing, DRC/ERC, Gerber & BOM Generation, Soldering & Board Bring-up
- **Protocols & Interfacing:** I2C, SPI, UART, GPIO, MQTT, LoRaWAN, ESP-NOW, HTTP/REST, WebSockets
- **Programming Languages:** C, Embedded C, C++, Python, Java, SQL, RISC-V Assembly
- **Cloud & IoT Platforms:** HiveMQ, ThingSpeak, Firebase, BlynkIoT, Google Cloud Platform (GCP)
- **Developer Tools & Lab Equipment:** Git, VS Code, PlatformIO, Linux/Bash, GDB, Oscilloscope, Logic Analyzer, Multimeter, JTAG/SWD, CST Microwave Studio, Keysight ADS

Featured Projects

Bare-Metal RISC-V Trap Handler & Context Switcher *ESP32-C6 / Bare-Metal*

Tech Stack: RISC-V Assembly, C, Direct MMIO, Hardware Timers, Interrupt Controller

- Architected a bare-metal trap handler and preemptive context switcher in assembly with 0-byte RTOS overhead, achieving ¡ 2.4 us context-switch latency across 32 registers
- Configured direct Memory-Mapped I/O (MMIO), machine timer interrupts (MTIE), and trap vectors (mtvec) for sub-microsecond hardware interrupt handling and register preservation

ESP-Drone – Autonomous Flight Controller & Firmware *ESP32-C6 / Quadcopter Firmware*

Tech Stack: C/C++, ESP-IDF, FreeRTOS, IMU Sensor Fusion, PID Control, ESP-NOW / Wi-Fi, MCPWM

- Engineered real-time flight control firmware on ESP32-C6 RISC-V architecture, achieving 500 Hz closed-loop attitude stabilization with ¡ 1% loop jitter
- Implemented 6-DOF IMU sensor fusion with complementary filtering, maintaining +-1 deg orientation accuracy, 20 kHz motor PWM, and ¡ 5 ms ESP-NOW telemetry

Dual-Axis Solar Tracker *Embedded Control System*

Tech Stack: Servo Motors, LDR Sensor Bridge, Closed-Loop Control, C/C++

- Constructed a closed-loop dual-axis solar tracking system, increasing photovoltaic energy capture by 28% over static panels through analog LDR differential sensing
- Fabricated motor driver and voltage regulation circuits; programmed dual-servo tracking algorithms across 180 deg azimuth and 90 deg elevation with low-power sleep modes

Digital Twin for Industrial Monitoring *5G-Integrated IoT System*

Tech Stack: ESP32, Sensor Fusion, 5G Telemetry, Python, Real-Time Dashboard

- Architected an industrial digital twin connecting multi-sensor ESP32 edge nodes with 5G cellular telemetry, sustaining 99.8% data transmission uptime
- Engineered a sub-100 ms edge-to-cloud sensor ingestion pipeline and real-time dashboard visualization for predictive vibration and thermal anomaly detection

Conformal Helmet-Integrated Antenna Array *RF & Defense Hardware*

Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band, Flexible PCB Substrate, CST Studio, ADS

- Engineered a flexible microstrip patch antenna array conformal to curved helmet geometry, achieving 45% profile reduction and S11 return loss ¡ -10 dB across UHF/L-band
- Modeled 3D radiation patterns and gain in CST Microwave Studio and Keysight ADS, integrating multi-layer copper-clad polyimide with high-isolation RF shielding

Leadership & Experience

Executive Technical Organizer 2024 – Present
IETE Students' Forum (ISF) *Chennai, India*

- Spearheaded technical hackathons and hands-on embedded systems workshops, mentoring 500+ student engineers across firmware development, sensor interfacing, and MCU bring-up
- Conceptualized, engineered, and deployed 10+ embedded and IoT systems end-to-end, spanning custom PCB bring-up, bare-metal firmware, and real-time cloud dashboards